Probability and statistics lectures (PHYS 6350)

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Abstract

The following notes are based on material from Boas (Chapter 15), Mathews and Walker (Chapter 14), Sec III of Living Reviews in Relativity article by Romano and Cornish

1 Suggested problems

From Boas, Chpt 15: 1.2, 1.5, 2.13, 3.3, 3.5, 3.18, 3.19, 4.10, 5.9, 5.11, 5.13, 5.16, 6.13, 6.14, 6.15, 6.16, 7.9, 7.12, 7.13, 8.8, 9.1, 9.2, 9.4, 9.7, 9.10, 10.1, 10.2, 10.3

2 Motivation

- Mathematics is based on *deduction*: one starts with a set of postulates (assumed to be true) and then derives the valid consequences of these postulates using the rules of logic.
- Mathematical statements are either true or false, with no uncertainty.
- Which postulates to start with is the creative aspect of mathematics.
- Example: Euclidean (flat space) vs non-Euclidean (curved space) geometry depends on whether or not we include the postulate that "two lines that are initially parallel never intersect and never diverge".
- Science, on the other hand, is based on *induction*, trying to make general statements about the state of nature from observations, which are *incomplete* and *imprecise* (due to measurement noise). Thus, our conclusions about nature are necessarily uncertain.
- Probabilistic inference (aka "plausible inference", "statistical inference") is the way that science deals with uncertainty.

3 Preliminaries

- Start by considering games of chance: flipping coins, rolling die, drawing cards from a deck, selecting colored balls from a bag, ... These are all examples of "repeatable, identical experiments", which is needed for the "classical" (or frequentist) interpretation of probability.
- According to the frequentist definition, probabilities cannot be assigned to hypotheses or parameters describing the state of nature, which have fixed but unknown values.
- NOTE: A Bayesian has a more general definition of probability, interpreting it as "degree of belief" (or confidence, plausibility) in any proposition (need not be associated with repeatable identical experiments). So a Bayesian can assign probabilities to hypotheses or parameters describing the state of nature.
- In what follows, we will restrict attention to the frequentist interpretation of probability, but note that all of the following results apply as well to the more general Bayesian interpretation.

- An example of an "experiment" would be rolling two die and recording the numbers on the the two faces of the die.
- For such experiments, the probability of an event A is defined as the long-run relative frequency of A occurring in a set of repeatable, identical experiments:

$$P(A) \equiv \lim_{N\to\infty} N(A)/N \tag{1}$$

  Here $N(A)$ is the number of times that the event A occurred and N is the total number of times that the experiment was performed.
- Sample space: set of *mutually-exclusive*, *collectively-exhaustive* outcomes of an experiment.
- Mutually-exclusive means that outcome A and B do not occur simultaneously (i.e., the joint probability of A and B is zero). Collectively-exhaustive means all possible outcomes are accounted for.
- Examples: for two successive flips of a coin, we can consider two different sample spaces:

$$S_1 \equiv \{tt, th, ht, hh\}\,, \qquad S_2 \equiv \{0h, 1h, 2h\} \tag{2}$$

  where $0h = tt$, $1h = \{th, ht\}$, $2h = hh$.
- Events in S_1 are equally likely (an example of a *uniform* sample space), while the events in S_2 have different probabilities (non-uniform sample space):

$$P(A \in S_1) = \{0.25, 0.25, 0.25, 0.25\}\,, \qquad P(A \in S_2) = \{0.25, 0.50, 0.25\} \tag{3}$$

3.1 Algebra of probability

- Probabilities satisfy the following:

  (i) Possible values:

$$0 \le P(A) \le 1 \tag{4}$$

  with 1 or 0 if A is certain to occur or certain not to occur.

  (ii) Sum rule:

$$P(A) + P(\bar{A}) = 1 \tag{5}$$

  where $\bar{A}$ is the complement of A (i.e., all outcomes different from A, with no overlap).

  (iii) Product rule:

$$P(A, B) = P(A|B)P(B) \tag{6}$$

  where $P(A, B)\,(= P(B, A))$ is the *joint probability* for both A and B to occur, and $P(A|B)$ is the *conditional probability* for A to occur assuming that B occurs.
- Note that $P(B|A) \neq P(A|B)$.
- Example: $A =$ "person is a woman"; $B =$ "person is pregnant". Then $P(B|A) \approx 0.30$, but $P(A|B) = 1$.
- If the conditional probability $P(A|B)$ is independent of B (i.e., $P(A|B) = P(A)$), then A and B are said to be *statistically independent*.
- It follows that if A and B are statistically independent then

$$P(A, B) = P(A)P(B) \tag{7}$$

- If A and B are *mutually exclusive*, then

$$P(A, B) = 0 \tag{8}$$

- Define $P(A+B)$ to be the probability that "events A or B or both A and B occur". Then

$$P(A+B) = P(A) + P(B) - P(A,B) \tag{9}$$

- Example: Consider a sample space for which events A and B both occupy $2/3$ of the total area, with $1/3$ overlap. In terms of probabilities, we have

$$P(A) = P(B) = 2/3\,, \quad P(A,B) = 1/3\,, \quad P(A|B) = P(B|A) = 1/2 \tag{10}$$

Note also that

$$P(A,B) = P(A|B)P(B) = P(B|A)P(A)\,, \quad P(A+B) = P(A) + P(B) - P(A,B) = 1 \tag{11}$$

3.2 Bayes' theorem

- Bayes' theorem:

$$P(B|A)P(A) = P(A|B)P(B) \tag{12}$$

or, equivalently,

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} \tag{13}$$

- Bayes' theorem is an immediate consequence of $P(A,B) = P(B,A)$.

- Applications of Bayes' theorem:

Exercise: Suppose on your last visit to the doctor's office you took a test for some rare disease. This type of disease occurs in only 1 out of 10,000 people, as determined by a random sample of the population. You are told that the test you took is rather effective in that it can correctly identify the presence of the disease 95% of the time, but it gives false positive 1% of the time.

Suppose the test came up positive. What is the probability that you have the disease?

Solution:

Let H = have the disease; $+$ = test positive

Information:

$$P(H) = 0.0001\,, \quad P(\bar{H}) = 0.9999\,, \quad P(+|H) = 0.95\,, \quad P(+|\bar{H}) = 0.01\,, \tag{14}$$

Calculate

$$P(H|+) = \frac{P(+|H)P(H)}{P(+)} \tag{15}$$

where

$$P(+) = P(+|H)P(H) + P(+|\bar{H})P(\bar{H}) = 0.95 \times 0.0001 + 0.01 \times 0.9999 \approx 0.01 \tag{16}$$

Thus,

$$P(H|+) \approx \frac{0.95 \times 0.0001}{0.01} \approx 0.01 \tag{17}$$

so 1% chance that you have the disease given that you tested positive

- Exercise: (Boas, Chpt 15, Sec 3, Example 3):

A prelim exam is given to students before they take a course. You are told that 95% of the students pass the course, so 5% fail the course. You are also told that 96% of the students who pass the course passed the prelim; while 25% of the students who fail the course passed the prelim.

What is the probability that a student will pass the course if they fail the prelim?

Solution:

Let A = student fails prelim and B = student passes course.

From the above information, we have:

$$P(B) = 0.95\,, \quad P(\bar{B}) = 0.05\,, \quad P(\bar{A}|B) = 0.96\,, \quad P(\bar{A}|\bar{B}) = 0.25 \tag{18}$$

Using the sum rule, we can also conclude:

$$P(A|B) = 0.04\,, \quad P(A|\bar{B}) = 0.75 \tag{19}$$

Thus,

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} \tag{20}$$

where $P(A|B)$ and $P(B)$ are given above, and

$$P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B}) = 0.04 \times 0.95 + 0.75 \times 0.05 = 0.0755 \tag{21}$$

leading to $P(B|A) \approx 0.50$.

4 Permutations, combinations, ...

- The total number of permutations of n objects equals

$$P(n) \equiv n(n-1)\cdots 1 = n! \tag{22}$$

since there are n possible choices for the first object, $n-1$ choices for the second object, ...

- The number of permutations of n objects taken m at a time

$$P(n,m) \equiv n(n-1)\cdots(n-m+1) = \frac{n!}{(n-m)!} \tag{23}$$

- The number of *combinations* of n objects taken m at a time ($m \leq n$) (so ignoring order):

$$C(n,m) \equiv \frac{P(n,m)}{P(m)} = \frac{n!}{(n-m)!\,m!} \equiv \binom{n}{m} \tag{24}$$

- $C(n,m)$ are the standard binomial coefficients, which appear when expanding $(a+b)^n$, namely

$$(a+b)^n = \sum_{m=0}^{n} \binom{n}{m} a^{(n-m)}b^m = a^n + na^{(n-1)}b + \frac{1}{2!}n(n-1)a^{(n-2)}b^2 + \cdots + b^n \tag{25}$$

- The above ideas are useful in statistical mechanics, where we consider distributing m particles in n states. (The m particles can be thought of as "balls" that we need to distribute among n "boxes").

- Maxwell-Boltzmann statistics: Consider (classical) distinguishable particles, like gas molecules in a container. The total number of equally-likely arrangements of m such particles in n different states is simply

$$nnn\cdots n = n^m \tag{26}$$

since there are n choices for each particle.

- Fermi-Dirac statistics: Consider m identical particles, for which only one or zero such particles can occupy a state (i.e., Pauli exclusion principle for fermions). Then the total number of equally-likely arrangements of m such particles in n different states is simply

$$\binom{n}{m} = \frac{n!}{(n-m)!\,m!} = \frac{n(n-1)\cdots(n-m+1)}{m!} \tag{27}$$

because each particle can occupy only one state, and the order of the particles doesn't matter since the particles are identical.

- Bose-Einstein statistics: Consider m identical particles, for which more than one particles can occupy the same state (i.e., bosons). Then the total number of equally-likely arrangements of m such particles in n different states is

$$\binom{n+m-1}{m} = \frac{(n+m-1)!}{(n-1)!\,m!} = \frac{(n+m-1)(n+m-2)\cdots n}{m!} \tag{28}$$

  Note that in comparison with the Fermi-Dirac result, the total number of equally-likely arrangements is larger for bosons since the effective number of states has increased from n to $n+m-1$.

5 Probability distributions

- A random variable is a function on sample space, e.g., x is the number of heads obtained when flipping a coin n times or the sum of the number of two die that are rolled repeatedly.
- The different possible values x_i of a random variable x have different probabilities $P_i = P(x_i)$.
- Random variables can be either discrete or continuous.
- A continuous random variable x has a *probability distribution* or *probability density function* $p(x)$, which has dimension of 1/dimension of x. This means that

$$p(x_0)\,\mathrm{d}x = \mathrm{Prob}(x \in [x_0, x_0 + \mathrm{d}x]) \tag{29}$$

- More generally,

$$\int_a^b \mathrm{d}x\, p(x) = \mathrm{Prob}(a \le x \le b) \tag{30}$$

5.1 Mean, variance, etc.

- *Mean* of x:

$$\mu \equiv \sum_i x_i P(x_i) \qquad \text{or} \qquad \mu \equiv \int_{-\infty}^{\infty} \mathrm{d}x\, xp(x) \tag{31}$$

  also denoted by $\langle x \rangle$.

- *Variance* of x:

$$\mathrm{Var}(x) \equiv \sum_i (x_i - \mu)^2 P(x_i) \qquad \text{or} \qquad \mathrm{Var}(x) \equiv \int_{-\infty}^{\infty} \mathrm{d}x\, (x-\mu)^2 p(x) \tag{32}$$

  also denoted by σ^2. Note that $\sigma \equiv \sqrt{\sigma^2}$ is called the *standard deviation*.

- Exercise: Prove that

$$\sigma^2 = \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2 \tag{33}$$

- *Cumulative density function*:

$$F(x_i) \equiv \sum_{x_j \le x_i} P(x_j) \qquad \text{or} \qquad F(x) \equiv \int_{-\infty}^{x} \mathrm{d}y\, p(y) \tag{34}$$

- *nth-moment* of a probability distribution:

$$\langle x^n \rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, x^n p(x) \tag{35}$$

- *Characteristic function*:

$$\phi(k) \equiv \int_{-\infty}^{\infty} \mathrm{d}x\, e^{ikx} p(x) \tag{36}$$

which is just the Fourier transform of $p(x)$.

- Inverse relation:

$$p(x) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}k\, e^{-ikx} \phi(k) \tag{37}$$

which is the inverse Fourier transform.

- The characteristic function is a *moment-generating function*, which can be seen by expanding the rhs of the definition of $\phi(k)$ as a power series in k and then performing the integrals over x:

$$\phi(k) = \langle e^{ikx} \rangle = 1 + ik\langle x \rangle - \frac{k^2}{2!}\langle x^2 \rangle + \cdots + \frac{(ik)^n}{n!}\langle x^n \rangle + \ldots \tag{38}$$

This leads to

$$\langle x^n \rangle = (-i)^n \left. \frac{\mathrm{d}^n \phi}{\mathrm{d}k^n} \right|_{k=0} \tag{39}$$

5.2 Joint probability distributions

- Let x and y be continuous random variables with joint probability distribution $p(x, y)$. Then the probability distributions for x and y separately are given by the integrals

$$p_X(x) = \int_{-\infty}^{\infty} \mathrm{d}y\, p(x, y)\,, \quad p_Y(y) = \int_{-\infty}^{\infty} \mathrm{d}x\, p(x, y) \tag{40}$$

- The *covariance* of x and y is defined by

$$\mathrm{Cov}(x, y) \equiv \langle (x - \langle x \rangle)(y - \langle y \rangle) \rangle = \langle xy \rangle - \langle x \rangle \langle y \rangle \tag{41}$$

where

$$\langle f(x, y) \rangle \equiv \int_{-\infty}^{\infty} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}y\, f(x, y)\, p(x, y) \tag{42}$$

for any function $f(x, y)$.

- Exercise: Show that if x and y are statistically independent (i.e., $p(x, y) = p_X(x) p_Y(y)$), then $\mathrm{Cov}(x, y) = 0$.

- But note that the converse is not true. In other words, $\mathrm{Cov}(x, y) = 0$ does *not* imply in general that x and y are statistically independent.

- Example: (Boas, Chpt 15, problem 6.16) Consider $p(x, y) = N(3y^2 + \cos x)$ defined over the rectangle $x \in [-\pi/2, \pi/2]$, $y \in [-1, 1]$, where N is a normalization constant. It has $\mathrm{Cov}(x, y) = 0$, but $p(x, y) \neq p_X(x) p_Y(y)$. So $\mathrm{Cov}(x, y) = 0$ does not imply x and y are statistically independent in general.

- An exception to this last statement is for x and y described by a *multivariate Gaussian* distribution (which we will describe in the next section). For this case, $\mathrm{Cov}(x, y) = 0$ implies that x and y are statistically independent, described by a product of 1-dimensional Gaussian distributions for x and y separately.

- For convenience, denote the two random variables by x_1, x_2. Then the characteristic function for the joint probability distribution $p(x_1, x_2)$ is defined as

$$\phi(k_1, k_2) = \int \mathrm{d}x_1 \int \mathrm{d}x_2\, e^{i(k_1 x_1 + k_2 x_2)} p(x_1, x_2) \tag{43}$$

which is a 2-d Fourier transform of $p(x_1, x_2)$.

- Expanding the exponential on the rhs leads to

$$\phi(k_1, k_2) = 1 + i\left(k_1\langle x_1\rangle + k_2\langle x_2\rangle\right) - \frac{1}{2!}\left(k_1^2\langle x_1^2\rangle + k_2^2\langle x_2^2\rangle + 2k_1k_2\langle x_1x_2\rangle\right) + \cdots \tag{44}$$

  in terms of the moments of x_1 and x_2.

- It immediately follows from the above definition of $\phi(k_1, k_2)$ that if x_1 and x_2 are statistically independent, then $\phi(k_1, k_2)$ has the form of a product

$$\phi(k_1, k_2) = \phi_1(k_1)\phi_2(k_2) \tag{45}$$

- The above definitions and results applies to an arbitrary number of random variables $x_1, x_2, \cdots, x_n$.

- Consider a new random variable via $z \equiv f(x, y)$, for some function $f(x, y)$. Then the characteristic function $\phi_Z(k)$ of $z = f(x, y)$ is given by

$$\phi_Z(k) = \int_{-\infty}^{\infty} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}y\, e^{ikf(x,y)} p(x, y) \tag{46}$$

- Exercise: Using the above expression for $\phi_Z(k)$, show that the probability distribution $p_Z(z)$ for $z = f(x, y)$ is given by

$$p_Z(z) = \int_{-\infty}^{\infty} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}y\, \delta(z - f(x, y))\, p(x, y) \tag{47}$$

- Example: Assume that x and y are statistically-independent random variables having probability distribution $p_X(x)$ and $p_Y(y)$ and characteristic functions $\phi_X(k)$ and $\phi_Y(k)$. Define a new random variable

$$z \equiv x + y \tag{48}$$

  Show that the characteristic function $\phi_Z(k)$ for $z = x + y$ is the product of the characteristic functions for x and z:

$$\phi_Z(k) = \phi_X(k)\phi_Y(k) \tag{49}$$

- We will use the above result in the next section when we prove the *central limit theorem*—i.e., that the probability distribution of the average of n independent samples of a random variables x with probability distribution $p(x)$ is Gaussian in the limit $n \to \infty$, *independent* of the form of the probability distribution $p(x)$.

6 Gaussian distribution

- The familiar symmetric "bell-shaped" curve that people often draw is for a *Gaussian probability distribution*:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}} \tag{50}$$

- The above distribution is properly normalized and has mean and variance given by the parameters μ and σ^2 appearing in the definition of $p(x)$.

- Exercise: Prove those statements.

- The *mode* (correponding to the maximum $p(x_{\text{mode}})$) and *median* (corresponding to total probability equal to 50% below (or above) x_{median}) agree with the mean for a Gaussian probability distribution.

- Exercise: Prove that the characteristic function $\phi(k)$ of the Gaussian distribution is given by

$$\phi(k) = e^{ik\mu} e^{-k^2\sigma^2/2} \tag{51}$$

  Recall that we saw this earlier in the topic on Fourier transforms—namely, that the Fourier transform of a Gaussian is another Gaussian.

- Central-limit theorem:

  Consider n independent samples x_i of a random variable x described by *any* probability distribution $p(x)$. Let μ and σ^2 denote the mean and variance of the random variable x (and hence the mean and variance of each of the random samples x_i).

  Define the new random variable

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{52}$$

  which is just the arithmetic mean of the N independent samples of x. Given this definiton of $\bar{x}$, it follows that

$$\langle \bar{x} \rangle = \mu\,, \qquad \mathrm{Var}(\bar{x}) = \sigma^2/n \tag{53}$$

  The central limit theorem says then says that in the limit $n \to \infty$, the probability distribution of $\bar{x}$ is Gaussian-distributed with mean μ and variance σ^2/n as given above.

- Exercise: Prove the central limit theorem (see Mathews and Walker, Section 14.5).

  Proof: The prove begins by showing that the characteristic function $\phi_Y(k)$ of the random variable $y \equiv \bar{x} - \mu$ is equal to the nth power of $e^{-ik\mu/n}\phi_X(k/n)$, i.e.,

$$\phi_Y(k) = \left[e^{-ik\mu/n} \phi_X\left(k/n\right) \right]^n \tag{54}$$

  where $\phi_X(k)$ is the characteristic function of x (or each random sample x_i).

  It then follows that

$$\begin{aligned} e^{-ik\mu/n}\phi_X(k/n) &= \int \mathrm{d}x\, e^{ik(x-\mu)/n} p(x) \\ &= 1 + i\frac{k}{n}\langle x-\mu\rangle - \frac{1}{2}\frac{k^2}{n^2}\langle (x-\mu)^2\rangle + \cdots \\ &= 1 - \frac{1}{2}\frac{k^2}{n^2}\sigma^2 + \cdots \end{aligned} \tag{55}$$

  as a consequence of $\langle x - \mu \rangle = 0$ and $\langle (x-\mu)^2 \rangle = \sigma^2$.

  Raising this last expression to the nth power and taking the limit $n \to \infty$ yields

$$\phi_Y(k) = \left(1 - \frac{1}{2}\frac{k^2}{n^2}\sigma^2 + \cdots\right)^n \sim e^{-\frac{k^2\sigma^2}{2n}} \tag{56}$$

  where we used the following limiting expression for e^x:

$$\lim_{n\to\infty}\left(1 + \frac{x}{n}\right)^n = e^x \tag{57}$$

  Finally, since the characteristic function of a Gaussian is also a Gaussian, we can conclude that

$$p_Y(y) \sim \frac{1}{\sqrt{2\pi\sigma^2/n}} e^{-\frac{1}{2}\frac{y^2}{\sigma^2/n}} = \frac{1}{\sqrt{2\pi\sigma^2/n}} e^{-\frac{1}{2}\frac{(\bar{x}-\mu)^2}{\sigma^2/n}} \tag{58}$$

- Multi-variate Gaussian distribution:

  Consider two random variables x, y. Then the multivariate Gaussian distribution for x and y is

$$p(x) = \frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}d^T C^{-1} d} \tag{59}$$

  where

$$d = \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix}, \qquad C = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix} \tag{60}$$

  with C being the covariance matrix of x and y. Here, μ_x, μ_y and σ_x^2, σ_y^2 are the means and variances of x, y, respectively, and $0 \le \rho < 1$ is the *correlation coefficient* which enters in to expression for the covariance $\mathrm{Cov}(x, y)$ of x and y.

7 Binomial distribution

- Consider n flips of a fair coin, so that probability of heads for a single flip is $p = 1/2$ (which is also the probability of a tail). Then the probability of m heads out of n flips is given by:

$$P(m) = \binom{n}{m} \left(\frac{1}{2}\right)^m \left(\frac{1}{2}\right)^{(n-m)} = \binom{n}{m} \left(\frac{1}{2}\right)^n \tag{61}$$

- More generally, if you have n trials of some repeatable experiment, such that the probability for a "success" for each trial is p (so that the probability for "failure" is $q \equiv 1 - p$, then

$$P(m) = \binom{n}{m} p^m q^{(n-m)} = \binom{n}{m} p^m (1-p)^{(n-m)} \tag{62}$$

  is the probability of m "successes" out of n trials.

- The binomial distribution has its peak at or close to the value np, which the expected number of successes (which need not be an integer), i.e., $\langle m \rangle = np$.

- The binomial distribution also has variance $\sigma^2 \equiv \langle m^2 \rangle - \langle m \rangle^2 = np(1-p)$.

- Exercise: Prove these last two statements.

- There are two interesting limiting cases of the binomial distribution:

  (i) The limit $n \to \infty$ for fixed p (so that np also tends to infinity). In this limit

$$P(m) \sim \frac{1}{\sqrt{2\pi np(1-p)}} e^{-\frac{1}{2}\frac{(m-np)^2}{np(1-p)}} \tag{63}$$

  which is a Gaussian probability distribution with mean $\mu = np$ and variance $\sigma^2 = np(1-p)$

  Note the the width σ of this distribution relative its mean μ is given by

$$\frac{\sigma}{\mu} = \frac{\sqrt{np(1-p)}}{np} = \sqrt{\frac{1-p}{np}} \tag{64}$$

  which tends to zero as $n \to \infty$.

  (ii) The limit $n \to \infty$ and $p \to 0$ in such a way that np is fixed. In this limit

$$P(m) \sim \frac{(np)^m}{m!} e^{-np} \tag{65}$$

  which is the Poission distribution with $\mu \equiv np$ (see next section).

- Exercise: Prove (i) (see Boas, Chpt 15, Problem 8.8)

  This (rather long) proof requires the use of Stirling's approximation for factorials:

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n (1 + \mathcal{O}(1/n)) \tag{66}$$

  and replacing exponents like m and $n - m$ from the factorial approximations by

$$m = np + \delta = np\left(1 + \frac{\delta}{np}\right), \qquad n - m = nq - \delta = nq\left(1 - \frac{\delta}{nq}\right) \tag{67}$$

  where $q \equiv 1 - p$, and then making expansions in powers of δ/n. (See the solution for Boas, Chpt 15, Problem 8.8 for details.)

- Exercise: Prove (ii) (see Boas, Chpt 15, Problem 9.10)

  This makes use of two results:

$$\binom{n}{m} \equiv \frac{n!}{m!(n-m)!} \sim \frac{n^m}{m!}, \tag{68}$$

  and

$$p^m(1-p)^{(n-m)} = p^m(1-p)^n(1-p)^{-m} = p^m \left(1 - \frac{np}{n}\right)^n \left(1 - \frac{np}{n}\right)^{-m} \sim p^m e^{-np} \tag{69}$$

8 Poisson distribution

- As mentioned in the previous section, the Poisson distribution is given by

$$P(m) = \frac{\mu^m}{m!}\, e^{-\mu} \tag{70}$$

- Exercise: Prove that μ is the mean and variance of this distribution.
- One can interpret $P(m)$ as the probability of m events occurring in a unit time interval, where μ is the average number of events in that time interval.
- If you explicitly include the time interval Δt in the probability distribution, then

$$P(m) = \frac{(\lambda \Delta t)^m}{m!}\, e^{-(\lambda \Delta t)} \tag{71}$$

  where λ is the rate of occurrence of the events.

- Examples:

  (i) m is the number of cars that cross an intersection over a 1-minute interval.

  (ii) m is the number of decays of a radioactive substance every second.